Pre-Lab 03: Active Filters and Oscillators

ECE 1212: Electronic Circuit Design Lab

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Activity	**Student Name**
Circuit Construction	Stefan Zhang
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Report Writing	Stefan Zhang; Nick Zhu

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1) Pre-Lab (3A): Low-Pass Filters

1.1) Question 1: Transfer Function

The circuit is a second-order Sallen-Key low-pass filter. For an ideal non-inverting amplifier,

$$K = 1 + \frac{R_B}{R_A}.$$

Applying KCL at the two RC nodes,

$$\frac{V_1 - V_{in}}{R_1} + \frac{V_1 - V_2}{R_2} + sC_1(V_1 - V_{out}) = 0,$$

$$\frac{V_2 - V_1}{R_2} + sC_2V_2 = 0, \qquad V_{out} = KV_2.$$

Solving these equations gives

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{K}{R_1R_2C_1C_2s^2 + [C_2(R_1 + R_2) + R_1C_1(1 - K)]s + 1}.$$

Comparing with the standard second-order form,

$$H(s) = \frac{K\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2},$$

we obtain

$$\omega_0 = \frac{1}{\sqrt{R_1R_2C_1C_2}}, \qquad f_0 = \frac{\omega_0}{2\pi},$$

$$Q = \frac{\sqrt{R_1R_2C_1C_2}}{C_2(R_1 + R_2) + R_1C_1(1 - K)}.$$

1.2) Question 2: Filter 1 Design

The specifications are

$$R_1 = R_2, \quad C_1 = C_2, \quad f_0 = 380\,\text{Hz}, \quad Q = 0.707.$$

For equal resistors and capacitors,

$$f_0 = \frac{1}{2\pi RC}, \qquad Q = \frac{1}{3 - K}.$$

Therefore,

$$K = 3 - \frac{1}{0.707} \approx 1.586.$$

Choose

$$R_A = 10\,\text{k}\Omega, \qquad R_B = 5.9\,\text{k}\Omega,$$

giving

$$K = 1.59, \qquad Q = 0.7092.$$

The selected filter components are

$$R_1 = R_2 = 12.7\,\text{k}\Omega, \qquad C_1 = C_2 = 33\,\text{nF}.$$

The resulting frequency is

$$f_0 = \frac{1}{2\pi(12.7 \times 10^3)(33 \times 10^{-9})} \approx 379.75\,\text{Hz}.$$

The transfer function becomes

$$H_1(s) = \frac{1.59}{1.7564 \times 10^{-7}s^2 + 5.9093 \times 10^{-4}s + 1}.$$

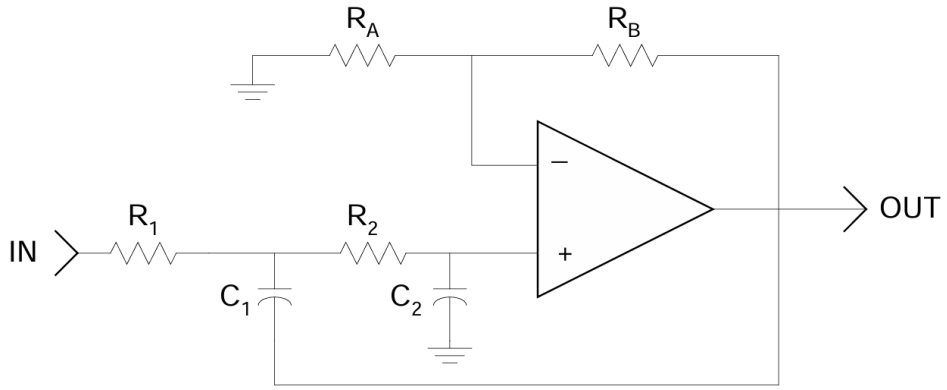


Figure 1: Designed Sallen-Key low-pass Filter 1

1.3) Question 3: Filter 2 Design

The specifications are

$$R_1 = R_2, \quad f_0 = 380\,\text{Hz}, \quad Q = 1,$$

with $R_B = 0$ and $R_A \to \infty$. Therefore, the op-amp is a voltage follower with $K = 1$.

For $R_1 = R_2 = R$,

$$Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}}.$$

To obtain $Q = 1$,

$$C_1 = 4C_2.$$

Choose

$$R_1 = R_2 = 17.8\,\text{k}\Omega,$$

$$C_1 = 47\,\text{nF}, \qquad C_2 = 12\,\text{nF}.$$

The actual design values are

$$Q = \frac{1}{2}\sqrt{\frac{47}{12}} \approx 0.9895,$$

$$f_0 = \frac{1}{2\pi R\sqrt{C_1 C_2}} \approx 376.50\,\text{Hz}.$$

The transfer function becomes

$$H_2(s) = \frac{1}{1.7870 \times 10^{-7} s^2 + 4.2720 \times 10^{-4} s + 1}.$$

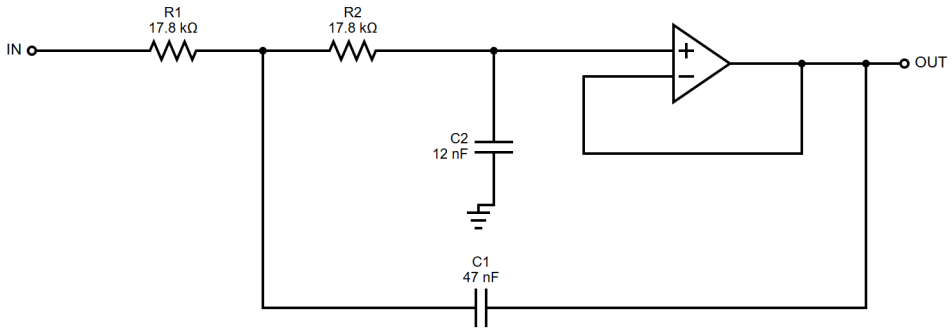


Figure 2: Designed Sallen-Key low-pass Filter 2

1.4) Question 4: Frequency Response and Pole-Zero Analysis

For a second-order low-pass filter, let $r = \omega/\omega_0$. The magnitude response is

$$|H(j\omega)| = \frac{K}{\sqrt{(1-r^2)^2 + (r/Q)^2}},$$

and the phase response is

$$\angle H(j\omega) = -\operatorname{atan2}\left(\frac{r}{Q}, 1 - r^2\right).$$

The theoretical magnitude and phase responses are plotted using MATLAB.

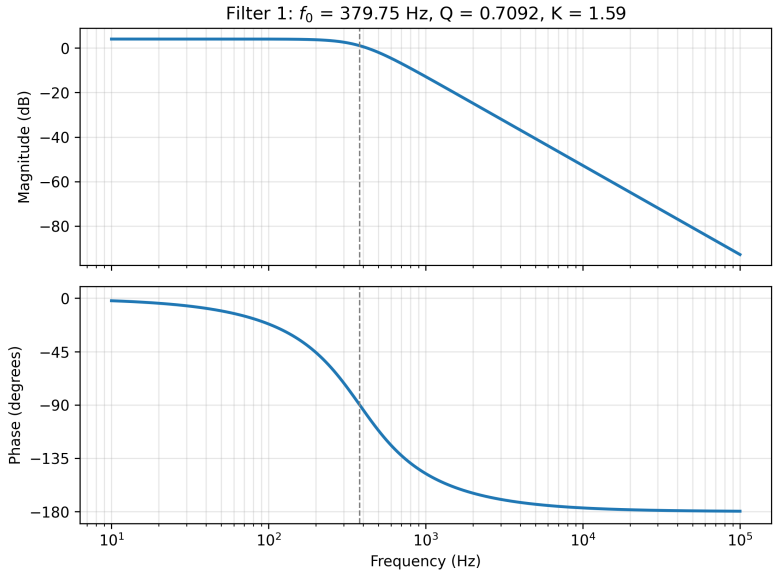


Figure 3: Theoretical magnitude and phase responses of Filter 1

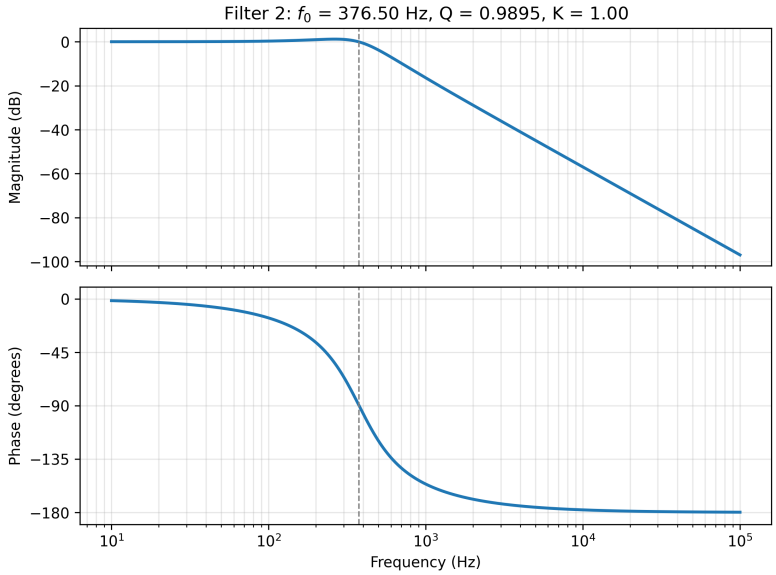


Figure 4: Theoretical magnitude and phase responses of Filter 2

The expected DC gains are

$$H_1(0) = 1.59, \qquad H_2(0) = 1.$$

The poles are

$$s_{1,2} = -\frac{\omega_0}{2Q} \pm j\omega_0\sqrt{1 - \frac{1}{4Q^2}}.$$

Both filters have two left-half-plane poles and no finite zeros, indicating stable responses.

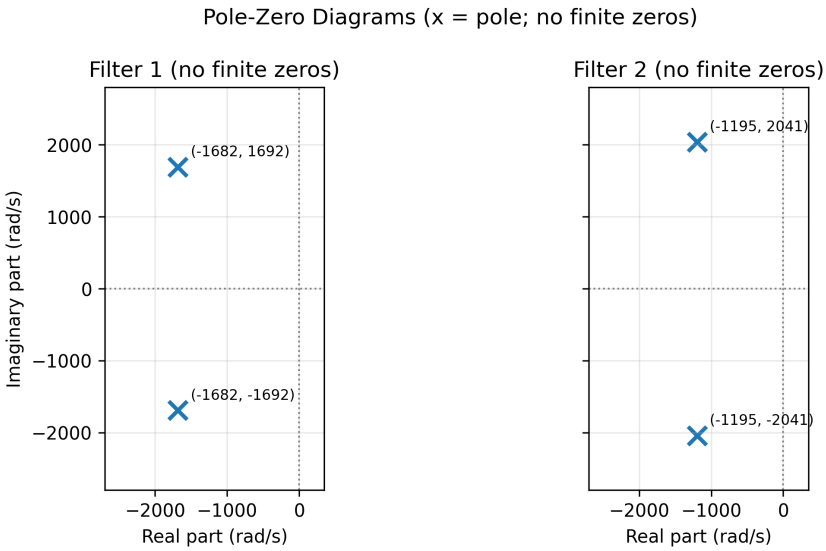


Figure 5: Pole-zero diagrams of Filter 1 and Filter 2

1.5) Question 5: Effect of Gain and Quality Factor

For Filter 1,

$$K = 1 + \frac{R_B}{R_A}, \qquad Q = \frac{1}{3 - K}.$$

Changing R_B by approximately $\pm 15\%$ produces the following theoretical results.

Table 1: Effect of feedback resistance on Filter 1

R_B (kΩ)	Gain K	Q
5.015 (-15%)	1.5015	0.6673
5.900 (Nominal)	1.5900	0.7092
6.785 (+15%)	1.6785	0.7567

Increasing R_B increases the gain and Q, resulting in stronger resonant peaking. The natural frequency remains unchanged in the ideal model.

For Filter 2,

$$Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}}.$$

Achieving $Q \geq 10$ requires

$$\frac{C_1}{C_2} \geq 400.$$

This large capacitance ratio complicates component selection and increases sensitivity to tolerances and op-amp nonidealities.

2) Pre-Lab (3B): Wien-Bridge Oscillator

2.1) Question 1: Oscillation Frequency

The Wien-bridge oscillator uses a frequency-selective positive-feedback network.

The specified component values are

$$R = 12\,\mathrm{k\Omega}, \qquad C = 0.01\,\mu\mathrm{F}.$$

The series and parallel RC impedances are

$$Z_s = R + \frac{1}{sC}, \qquad Z_p = \frac{R}{1 + sRC}.$$

The positive-feedback factor is

$$\beta(s) = \frac{Z_p}{Z_s + Z_p} = \frac{sRC}{s^2R^2C^2 + 3sRC + 1}.$$

According to the Barkhausen criterion,

$$|A\beta| = 1, \qquad \angle A\beta = 0^\circ.$$

For zero phase shift,

$$1 - (\omega_0 RC)^2 = 0.$$

Therefore,

$$\omega_0 = \frac{1}{RC}, \qquad f_0 = \frac{1}{2\pi RC}.$$

Substituting the specified values,

$$f_0 = \frac{1}{2\pi(12 \times 10^3)(0.01 \times 10^{-6})}$$

$$f_0 \approx 1326.29\,\mathrm{Hz} \approx 1.33\,\mathrm{kHz}.$$

2.2) Question 2: Minimum Gain for Oscillation

At the oscillation frequency,

$$\omega_0 RC = 1, \qquad \beta(j\omega_0) = \frac{1}{3}.$$

The Barkhausen magnitude condition gives

$$A\beta = 1 \quad \Rightarrow \quad A = 3.$$

For the non-inverting amplifier,

$$A = 1 + \frac{R_2}{R_1} = 3.$$

Hence, the critical resistance ratio is

$$\left(\frac{R_2}{R_1}\right)_{\text{critical}} = 2.$$

To start oscillations from noise, the amplifier gain should be slightly greater than 3:

$$\frac{R_2}{R_1} > 2.$$

For circuit construction, choose

$$R_1 = 10\,\text{k}\Omega,$$

$$R_2 = 18\,\text{k}\Omega + 5\,\text{k}\Omega \text{ potentiometer}.$$

This provides an adjustable resistance ratio from 1.8 to 2.3, covering the oscillation threshold.

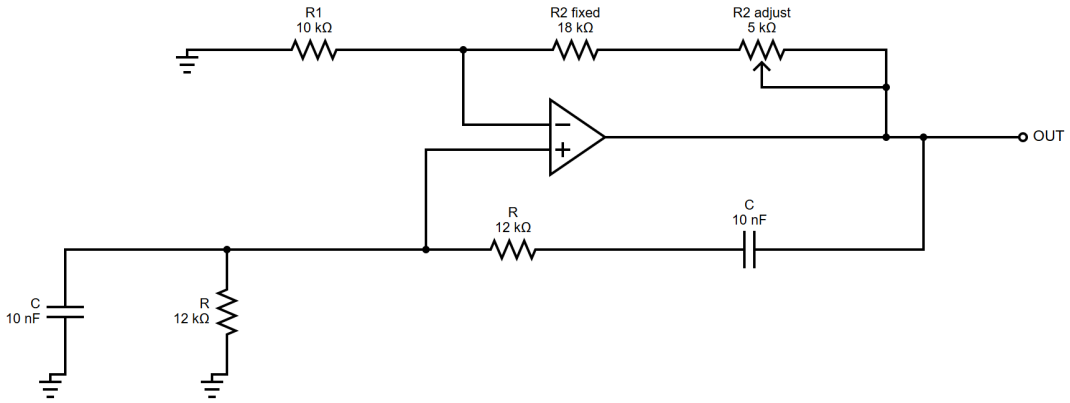


Figure 6: Wien-bridge oscillator with adjustable feedback gain

2.3) Question 3: Amplitude Stabilization

If the loop gain exceeds unity, the oscillation amplitude grows until limited by the op-amp output swing, causing waveform distortion.

An amplitude-stabilization circuit adjusts the effective amplifier gain as the output amplitude changes. It reduces the gain toward $A = 3$ as the amplitude increases, maintaining approximately unity loop gain and a stable sinusoidal output.